

Empowering Community Water Data Stakeholders

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Access to clean water is a critical challenge and opportunity for community-level collaboration. People rely on local water sources, but awareness of water quality and participation in water management is often limited. Lack of community engagement can increase risks of water catastrophes, such as those in Flint, Michigan, and Cape Town, South Africa. We investigated water quality practices in a watershed system serving c.100 000 people in the United States. We identified a range of entities including government and nonprofit citizen groups that gather water quality data. Many of these data are accessible in principle to citizens. However, the data are scattered and diverse; information infrastructures are primitive and not integrated. Water quality data and data practices are hidden in plain sight. Based on fieldwork, we consider sociotechnical courses of action, drawing on best practices in human-computer interaction and community informatics, data and environmental systems management.

RESEARCH HIGHLIGHTS

- We identify and investigate a water quality stakeholder network that is community-articulated and loosely coordinated.
- We characterize nested communities of practice in water quality monitoring, their interrelationships and their coordination work.
- We identify water quality monitoring as community work ‘hiding in plain sight’ and analyze the ramifications of this.
- We propose approaches to make water quality, and citizen-science water quality monitoring, more visible and accessible to community members.

Keywords: community data; citizen science; data literacy; informal learning; water quality

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1. INTRODUCTION

The *Lancet* reports that environmental pollution was responsible for 9 million premature deaths in 2015, accounting for 16% of all deaths worldwide (which is far more than from AIDS, tuberculosis and malaria combined); with ~20% of environmental pollution deaths attributable to water pollution (Landrigan *et al.*, 2017). The recent water crisis due to lead pollution in Flint, Michigan, is a vivid example (Pieper *et al.*, 2017). Climate change and rigidly optimistic legal and policy frameworks for water allocation have created intermittent shortage crises in many places, including Cape Town, Jakarta,

Melbourne, Mexico City, Sao Paulo, and, in the United States, from Texas to California (Welch, 2018).

Water pollution and water shortages are globally important, but they are also felt locally. In communities across America, for example, there is widespread public interest in water resources. Nonprofit citizen groups are engaged in water quality monitoring, assessment and data gathering. We identified a collection of stakeholder groups in a watershed system that serves ~100 000 people. Many of these groups have unique missions and distinctive membership, but they also coordinate through higher-order nonprofit groups and cross-municipality entities. The groups make use of various collaborative tools

(such as Google documents and social media) in order to maintain and cultivate datasets describing different aspects of water quality in the community. However, in spite of a general awareness that there are other groups and information, there is minimal visibility and access to datasets across groups. Access to more water quality data can provide a richer overall picture of the health and sustainability of a watershed, which could in turn strengthen community responses to land development issues. Today, new opportunities for supporting data-driven social interactions are developing, which demonstrate the value of a human-centered approach to the creation and use of community water data. Indeed, these might constitute an opportunity space for research in human-computer interaction (HCI) as data proliferate. This paper demonstrates the value of a human-centered approach to the creation and use of community water data.

We describe our project to understand a community watershed as a sociotechnical system. Through a series of interviews, we identified key stakeholder groups and their interactions. We investigated their water quality monitoring and data practices through interviews, hackathons and field observations. The activity of local water quality monitoring is serious community work; it safeguards a vital collective resource. It strengthens, leverages and balances water regulation through various levels of government. It represents the community. Yet it also appears loosely coordinated, informally supported and, to a considerable extent, invisible.

We characterize this effort as a community-articulated stakeholder network of citizens actively caring for fellow citizens through various water quality monitoring practices. These practices integrate citizen science, community engagement and health: the participants use scientific methods and instruments to track the quality of the water they, and their community, depend upon, and, in some cases, they do this by hiking into remote locations to obtain samples and measurements. The citizen science in this activity is interesting in that the citizens are in charge of the work; they are not merely carrying it out. The science is hyperlocal in that it is about a locale, takes place entirely within that locale and is produced and used by the people of the locale. The motivation for, experience of and result from this learning are more immediately consequential for participants than are typical in citizen science: monitoring the quality of local public water creates a local political resource to ensure the future quality of public water.

We engaged the stakeholder groups in workshops to demonstrate and describe their work and to envision a notional water quality data platform through which their data could be more accessible to the community. The collaborative work of community water data, including both the data practices of specific nonprofits, municipal and other governmental groups, and the coordination mechanisms among groups, is an intricate and critical collaborative system. This system embodies and is embedded in various rules and regulations that frame (and constrain) water quality data collection and analysis. Under-

standing and enhancing local sociotechnical systems for water monitoring is a specific new trajectory for leveraging and developing HCI and one that could help to address the global water crisis. More generally, the utilization of HCI concepts and techniques to leverage and support local water quality monitoring illustrates how HCI can be applied to other social and community level projects and challenges.

2. BACKGROUND

Our project draws upon several lines of prior research and innovation that inform community engagement with water data. First, we characterize water quality management as a vital public service that is sometimes challenged, but also innovative. Second, we draw upon citizen science research, which we discuss below, illustrating how, and with what effects, citizens participate in data gathering and analysis. Third, we draw upon recent work in community informatics characterizing innovation as a critical function and activity focus in contemporary communities. Finally, we draw upon work in informal learning as a focus for community-wide learning about environmental issues as well as data visualization and analysis.

2.1. Water quality management

Water quality management is a core public service of government. Citizens need to be confident that their local water supplies are adequate and safe. In the United States, for example, the Environmental Protection Agency (EPA) was created in 1970 to coordinate research, monitoring, standard-setting and enforcement activities. This effort includes a wide range of possible threats and types of pollution, but, in many cases, what is threatened or contaminated is public water.

In the past several decades, people throughout the world have become more aware of water resources as those resources have deteriorated. The watershed system that is the site for our research (described more below) was and still is well-known for trout fishery, but the region has faced continuing water quality challenges, including agricultural runoff, stormwater silt and invasive species. It has legacy pollution sources, including abandoned clay and coal mines and a chemical plant that, from 1982–2017, was federally designated as a contaminated site (an EPA *superfund* site). This watershed is not an extreme example. Rather, it indicates the level of effort and attention required to protect water resources in a complex society.

Monitoring and managing a water system is challenging, but failing to do so is potentially disastrous. For example, in 2014, public water sourcing in Flint, Michigan, was reconfigured in a way that caused lead to leach into the water delivered to local homes and businesses. Citizens consumed contaminated water for years until this situation came to light. In another example, public water sources in Cape Town, South Africa, were forecast to be completely depleted by November 2017 due to climatic perturbations in moisture corridors. By late 2018, it appeared

that water supplies had recharged, but the longer-term future for Cape Town's water supply is uncertain.

Partly in response to such challenges, citizens have started to become more aware and energized about water quality and water quality monitoring. A wide variety of citizen water groups have emerged. For example, in the state of Pennsylvania, in the United States, there are ~350 community watershed groups and associations addressing aspects of water quality and its implications (POWR, 2019). These groups make counts for key indicators of water system vitality, such as fish nests and macro-invertebrate insects; measure key parameters, such as pH and stream velocity and volume; gather samples to be tested for various contaminants; and work on mitigation strategies, such as helping to create riparian buffers to increase shade and control erosion near streams. A recent Proceedings of the United States National Academy of Science (PNAS) study found that citizen participation in water quality monitoring correlates with quality of public water supplies (Grant & Langpap, 2018; NWQMC, 2018).

Increasing public interest in water quality data is also encouraging the emergence of various kinds of software tools to handle data. Thus, the open data posted publicly by government agencies, like the US EPA (<https://www.epa.gov/waterdata>) and the US Geological Survey (<https://waterdata.usgs.gov>), have motivated development of visualization and analysis tools on the agency websites. This has encouraged the development of research platform projects, such as the hydrological information tools of the Consortium of Universities for the Advancement of Hydrologic Science (<https://www.cuahsi.org>) and the Stroud Water Research Center (<https://stroudcenter.org>). It has also encouraged the development of commercial data platforms, such the Aquarius product of Aquatic Informatics (<https://aquaticinformatics.com>). Despite the availability of such tools, many citizen groups still rely primarily on fairly primitive and generic information infrastructures, such as Google Sheets.

2.2. Citizen science

Citizen science is a category of civic innovation in which volunteers contribute to scientific research projects, and it has been a productive topic for HCI researchers (Law *et al.*, 2017). Citizen science projects can involve a variety of scientific activities, from image classification to audio-recording in the field, and they often involve collecting data from a wide area or over long period of time (Cooper *et al.*, 2010; Hand, 2010). Scientists benefit by distributing the work of data gathering; citizen participants benefit by learning about science—the process of conducting scientific research, relevant scientific terminology (Harandi *et al.*, 2018) and the values that guide such research projects (Newman *et al.*, 2012; Bonney *et al.*, 2009). Recent HCI research has attended to effective strategies for recruiting newcomers to citizen science projects (Lee *et al.*, 2017) as

well as how their participation may develop over time (Jackson *et al.*, 2016).

Depending on the project, participation can look quite different. Some projects rely on participants to collect and contribute data (e.g. air quality, traffic and noise). In one famous example, citizen scientists played an online protein folding game in order to improve a model enzyme (Cooper *et al.*, 2010). The results of their gameplay produced a scientific innovation culminating in a publication in the journal *Nature*. The scale and diversity of the teams that undertake citizen science projects can be vast. How different kinds of participation, cultural norms and institutional structures influence citizens' motivation to contribute to citizen science project is a particularly relevant topic for HCI researchers (Rotman *et al.*, 2014).

The citizen science activity we describe in this paper is a kind of 'extreme citizen science' (Burgess *et al.*, 2017; Kim *et al.*, 2011): the citizens are not only participants, they also plan and manage the activity. Extreme citizen science is 'a situated, bottom-up practice that takes into account local needs, practices and culture and works with broad networks of people to design and build new devices and knowledge creation processes' (ExCiteS, n.d.). In our study, citizens direct and execute the projects; indeed, ownership of their practices seems to be an important motivational dynamic. The resulting data collected through this practice are also used by local community groups in various ways (described in further sections below). This level of responsibility-taking allows and requires more innovation and raises research questions about possible barriers and facilitators to citizen science in water quality. For example, the citizens themselves are responsible for coordinating this work and for disseminating it to the larger community. We describe this type of extreme citizen science as *hyperlocal*, meaning it is *about* a locality, *planned and carried out* by people of the locality and has its primary *uses and impacts* locally.

2.3. Community engagement and development

Community is construed as a source of social support and as a foundation for civil society. Communities are part of, and reflect, the broader contemporary social context (Le Dantec, 2012). Contemporary communities are not just about identity, trust and mutual support, they are also about human development and the cultivation of creativity and innovation. Thus, it is normal for community life to involve learning and skill development. The concept of community was born out of Tönnies (1887) analysis of how technological disruptions of the 19th century threatened a traditional social order. This now-traditional meme of 'lost', or disappearing, community has helped orient significant critical analysis of technologies from power looms to televisions. Fortunately, contemporary community is, at the same time, a uniquely supportive context for citizen learning and innovation, and should be analyzed as constructive, creative and empowering (Carroll, 2012; Carroll *et al.*, 2015b).

People experience a sense of community and build social capital when they engage in reciprocal interactions based on time spent sharing, or instances of sharing, in contrast to feelings of isolation and withdrawal that often accompany the use of money (Carroll & Bellotti, 2015; Han *et al.*, 2014). ‘Habitat for Humanity’ and ‘Meals on Wheels’ illustrate the kinds of reciprocal interactions that lead people to experience a sense of community and build social capital. To our knowledge, water quality monitoring has not been framed as an analogous type of activity. However, it is clear from our engagement with the local water quality community that such a framing is appropriate and important.

Water quality monitoring can be understood as a critical engagement with the community. It can benefit people who do it in analogous ways, for example, by cultivating their knowledge of water resources and management, which, in turn, better equips them to participate in civic life where these issues are of central concern. These issues could have to do with proposed land rezoning and development or the proposed building of water bottling plant and how it might affect local water resources. It also expands the concept of how local nonprofits strengthen community health, including trust. Moreover, our case illustrates new sources and opportunities for innovation and learning throughout the community. For example, the water-monitoring groups are involved in school programs to help local children become more informed and involved with water resources. The groups have collaborated on ambitious and innovation informal learning programs to broaden awareness and participation in the wider community.

Human organizations, including communities, also experience distinct challenges. One that has been widely investigated in HCI is ‘invisibility’, as in invisible work (Suchman, 1995). Work can be critical to a group or organization, and yet not be acknowledged. Suchman discussed the critical contributions of legal assistants in a law firm to the overall flow of work activity that were nonetheless often invisible to firm’s partners. When we first ‘discovered’ the water quality groups, we realized that, up that moment, they were largely invisible to us, even though their driving mission is to ensure our water supply.

2.4. Informal learning

In 2018 Cape Town, South Africa, faced the prospect of shutting down its municipal water supply. The day of the shut-down was referred to as ‘Day Zero’. In the United States, most public water supplies in the Southwest are among the most at-risk systems in the world (Welch, 2018). The water crisis in Flint, Michigan, USA, was a particularly egregious case of water system mismanagement. It would be better for public institutions to monitor and maintain water resources and infrastructure responsibly, but this does not always happen. An effective strategy to reconfigure watershed management, and strengthen citizen participation in the process (Vlachokyriakos *et al.*, 2016; Harding *et al.*, 2015), could be citizen engagement

in informal learning in this consequential and concrete problem domain, in authentic learning contexts (Savery & Duffy, 1996; Lave & Wenger, 1991).

Authentic learning incorporates artifacts and materials that support and embody situated skills (Vygotsky, 1978). Such contexts integrate collaboration, social norms, tool manipulation, domain-specific goals and heuristics, problem solving and reflection-in-action directed at real-world challenges, such as water quality and land conservation. They enable better learning experiences and better outcomes (Dewey, 1966; Glassman, 2001; Palincsar, 1998; Suthers & Hundhausen, 2001; Vygotsky, 1978). For example, local community water quality monitors learn about water resources, water chemistry and the insects and fish that can thrive in clean water. They then apply this knowledge in public fora, including community meetings designed to solicit input from residents about land rezoning and development initiatives. Arguably, participation informed by an evidence base of local water data have the potential to be more meaningful, and impactful, than other kinds of participation. Such a community-based informal learning mechanism can be self-sustaining once it is established.

Parts of the broader local community are already engaged by our partner groups some of whom develop and implement instructional curricula at local middle and high schools. Our assumption is that broadening the availability and reach of informal learning opportunities could be useful for community members who care about an issue, intuitively recognize its importance and want to become more informed and engaged. Water quality data could thus become more integral to how community members present themselves (e.g. in public fora and hearings) to higher levels of government and other local institutions.

As we developed our relationships and investigation of the water quality groups, we were impressed by their concern with sharing data more effectively among themselves and with the larger community. Sharing data is a sort of linchpin: it is a key challenge for the citizen science of local water quality; it enhances the potential community impact and visibility of water quality monitoring; and it is required to enable community-wide informal learning and effective protection of local water quality. We framed this concretely as a *notional water data platform*: an accessible community data infrastructure that could federate the data gathered by the various water quality groups and perhaps diverse other community-based data.

3. METHODOLOGY

We conducted interviews, field observations and hackathons with citizens who participate in data collection through one or more of various community-based organizations involved in local water quality monitoring. A snowball sampling approach was used to recruit participants. In total, we engaged

with 13 groups, including ClearWater Conservancy (CWC), Center County Pennsylvania Senior Environmental Corps. (CCPaSEC), Spring Creek Watershed Association (SCWA), Spring Creek Watershed Atlas (Atlas), Water Resources Monitoring Project (WRMP), Nittany Valley Environmental Coalition (NVEC), Pennsylvania Fish & Boat Authority (PA-FBA) and Trout Unlimited (TU). We also engaged with Penn State University units collecting water quantity and quality data, including the Penn State Office of Physical Plant (OPP) and the State Water Resources Research Center. Names of organizations, towns, municipalities and so forth have been changed to preserve anonymity. Appendix A.1. gives the distribution of participants across interviews and hackathons.

3.1. Interviews

We conducted 15 semi-structured interviews focused on understanding how and why local water quality groups operate as they do, who they work with and how, what constraints and limitations they face, and opportunities they see for improving the management of the watershed. The interviews also focused on understanding the different groups' practices and their interest and methods of public outreach. We were interested to understand how this network could impact the broader community through informal learning about the watershed and overall community engagement and participation.

3.2. Hackathons

We hosted three hackathons in order to gain insights from groups of stakeholders who have different experiences and perspectives on water quality monitoring in the Spring Creek Watershed. Hackathons can be construed as a specific kind of workshop focused on identifying and developing innovative outcomes (Taylor & Clarke, 2018; Trainer *et al.*, 2016). The term hackathon typically connotes playful and intense software improvisation, but recently it has been extended to other design domains. An event must entail a time-bounded and collaborative setting in which a team or teams work together under self-imposed pressure in order to be considered a hackathon. We were influenced by civic innovation hackathons (cf. Johnson & Robinson, 2014; Morelli *et al.*, 2017; Porter *et al.*, 2017) for this project. No software improvisation was involved.

These hackathons entailed discussions of challenges facing the groups and the watershed as well as possible innovations. We used the concept of a water quality data platform to probe participants about integrating and sharing water data, including the possibility of sharing data and visualization tools with local citizens and to cultivate broader interest and participation in water quality issues. We created a clickable prototype web application to probe hackathon participants. The prototype included (1) an interactive map showing monitoring sites around the local watershed; (2) a set of filtering tools for local water quality data collected by different stakeholder groups;

(3) a message board where citizens could post and answer questions about water quality and water resource management; and (4) an information board about existing water groups in the community. Our ambition was to collaborate with participants to envision an application that supported data access to a large, one-stop water data repository for the Spring Creek Watershed, make visible the associated metadata and evoke from the participants some of their work practices that could possibly be supported by such a data platform.

In the process, we also got a grasp of some interactions that are beyond the scope of a water data platform for the watershed (e.g. expert opinions on a public-facing application). As yet, there is not a system that supports all of these activities. In addition, we engaged stakeholders from different groups in discussions about the members of the local water quality network and how they interact with one another. Finally, hackathons provided opportunities to identify community-level issues, while interviews and field observations tend to focus on issues at the group level. We audio recorded and transcribed segments of each hackathon and, furthermore, collected artifacts produced by participants (e.g. sketches of the water quality network).

3.3. Field observations

We were interested in observing stakeholders who manually collect water samples in the field. Two CCPaSEC citizen groups responded to our requests to observe and record data collection activities in the field. Two researchers, who are also authors on this paper, made two separate trips into the field with the different citizen groups. They worked as participant observers—recording audio segments, writing field notes and taking digital photographs of activities, while also talking with participants about their experiences, asking questions and providing support to stakeholders (e.g. one water quality stakeholder asked the researchers for the time so that they could record it on a sheet of paper summarizing their collection activities). Some digital images of these field observations are reproduced in this paper Figs 1 and 2. These observations provided insight into a typical trip into the field for CCPaSEC citizen groups. Each group collected water samples, performed chemical tests and recorded their data. We observed team dynamics as well as the kinds of tools and technology they use to collect and record data.

3.4. Data analysis

All interviews and hackathon sessions were audio recorded with the consent of the participants. These recordings were transcribed and open-coded using the grounded theory method (Charmaz, 2014). In addition, researchers took detailed notes during field observations, which were also open-coded for analysis. A grounded theory approach to data analysis was chosen due to the exploratory nature of this project. Grounded theory



FIGURE 1. Volunteers in the field collecting water samples.



FIGURE 2. Analyzing samples at a stream site.

helps in providing theoretical explanation of social processes studied in a specific context (Glaser & Strauss, 1967). In this study, researchers were interested in understanding the social–technical processes of water quality monitoring as community engagement.

4. RESULTS: WATERSHED AS A SOCIOTECHNICAL SYSTEM

In this section, we describe the different community water groups, what they do and why, as well as how they interconnect. These community groups collect and test water samples, organize data sets and document collections, and share data with local and state government as well as with community members and groups. In addition, they enumerate threats to local watersheds, carry out watershed protection and restoration activities, and organize and participate in local meetings. We organize this section around three main results: (1) the water quality network is community-articulated and loosely coordinated; (2) Each group in the network has unique water quality community of practices (CoPs) that is further encapsulated in an integrated CoP for the entire watershed system; and (3) water quality data are not effectively visible to the public.

4.1. The water quality network is community-articulated and loosely coordinated

From our fieldwork and workshops, we observed how these groups collaborate with one another and some goals of their collaboration. There are two kinds of collaborations among different groups: formal and informal. ‘Formal collaboration’ describes established and accepted data sharing practices or financial support between groups. For instance, the WRMP gets financial support from several other organizations and local municipalities. At the same time, they host water quality data and distribute data to anyone who makes a request.

‘Informal collaboration’ describes cases where stakeholders voluntarily participate in different groups and serve as unsanctioned conduits for information. That is, one group does not necessarily encourage a stakeholder to participate and share information with another. However, this sort of sharing seems to be a natural occurrence when stakeholders participate with multiple groups. One participant described how she got involved in activities of multiple groups, ‘Well, I just wanted to know more [about CCPaSEC] once I got involved with Trout Unlimited, so everything I’ve done is from the base of Trout Unlimited and the chapter and then kind of trying to find out more about what goes on in this watershed... [and] in terms of my desire to have more knowledge.’ (P6)

Informal collaborations may stem from the fact that the research site is a hyperlocal community. Many of the same people engage with different groups and ‘wear different hats’. One participant referred to a collection of water quality

people as ‘the usual suspects’, for example. The water quality stakeholder groups are highly collaborative, but this does not mean that there is no sense of competition between groups. At one of our first hackathons, for example, the issues of quality assurance and quality control (QAQC) came up. QAQC processes look different for each group. Some groups send their data to sanctioned water testing laboratories for validation, and other groups have their own internal validation processes. Each group has a somewhat unique stake in the watershed and, thus, regard one another as engaged in similar but distinct activities and having similar but distinct practices. Issues like data quality evoke critical discussion and reflection. *How are data collection practices rigorous? What data are most reliable? How can data inform land rezoning and development decisions?* These sorts of questions seemed to provide motivation to participate in the water quality network in a way that is beneficial for all.

Whether the collaboration is formal or informal, coordination between groups is loose, which means that there are no firmly established protocols for collaboration or coordination. One participant described this notion of a community articulated and loosely knit structure as follows: ‘... *Spring creek watershed association is nothing more than a loosely knit group of grassroots people... we realized this is a pretty good group of people to be really protecting the watershed... We started going regularly (P2).*’ Hackathons yielded the insight that data requests from one group to another happen on an ad hoc basis. On the other hand, we also learned that some groups store data in open access repositories, which means that anyone who knows this, or who visits the group’s website, can access their data without submitting formal requests. Thus, data sharing can happen both formally and informally. There are no community-wide arrangements dictating what data to share or how to share data. Moreover, if and when volunteers participate in multiple groups, there are no requirements or expectations that they function in an official capacity to coordinate efforts between groups. Such coordination may happen, but it most likely happens on an ‘as needed’ basis.

Figure 3 illustrates a network of stakeholders that was created by a participant during a hackathon. This network is community-articulated in the sense that it has been defined by its members and not by any existing regime of regulation.

4.2. The water quality network is a community of practice made up of communities of practice

The water quality network in the Spring Creek Watershed is itself a CoP. Wenger (2011) describes CoPs as comprised of individuals who are passionate about the work they do, who are committed to the domain and who have formulated a way of doing work and developing shared resources for learning and engagement. This CoP is exhibited in several ways. As described in the previous section, the water quality network has formal and informal collaboration with respect to water

quality data. In collaboration with the SCWA, CWC maintains a database of collected information under the auspices of the WRMP. WRMP is a forum for professionals in the Spring Creek Watershed to provide leadership to the monitoring initiative. It also spurs exchange of ideas and future directions for the entire watershed. In one hackathon, participants referred to established processes for quality assurance and quality control. They also expressed mutual interest in making data accessible to one another and the broader community.

Each group within the network could also be construed as a discrete CoP given that each has unique methods and tools for collecting, analyzing and sharing water quality data. In the early stages of our work, interviewees were interested in learning about other groups’ practices, and during our first hackathon, representatives from different groups shared and discussed different data validation techniques. For some groups, third party validation is a crucial step in their process. For others, it does not matter.

The combination of interviews and hackathons revealed two distinct practices—one in the regulatory context and the second one, which was developed, managed and owned by community members. These practices differ in terms of the overall goals of the CoP and the actual data collection practices. Groups operating within the regulatory context like University Area Joint Authority (UAJA) (municipal authority providing wastewater treatment) and State College Borough Water Authority (SCBWA) use metadata and protocol for data collection that is established by higher federal and state organizations like the State EPA. The SCBWA collects and analyzes data to primarily demonstrate that the area’s drinking water meets federal and state-level water quality standards. There are other uses as well, e.g. informing the public about drinking water quality. The groups within the regulatory context maintain real-time data.

In contrast, the community led practices are directed toward data-driven local policy and decision-making. CWC maintains water-monitoring sites and a database of the measures at these sites in collaboration with SCWA to keep a record of the health of the streams. As their 2018 annual report states, ‘Data is used in making critical water use and planning decisions in the watershed’ (ClearWater Conservancy, 2018). Another participant provided a vivid description of data use: ‘... *Organizations like, even mining companies will ask—if they have a company on the watershed—they will ask for the water quality data. Usually temperature is what they look at mostly. [The local] sewer authority is interested in flow, temperature because they make beneficial reuse water and so they want to monitor where flow increases might be due to that or potential temperature increases using that as well (P1).*’

Community water groups like the CCPaSEC and TU have created their own data collection protocols based on their goal of water quality testing. For example, CCPaSEC collects data on temperature, stream flow, pH, conductivity, nitrates, dissolved oxygen, sulfates and phosphates monthly. They conduct

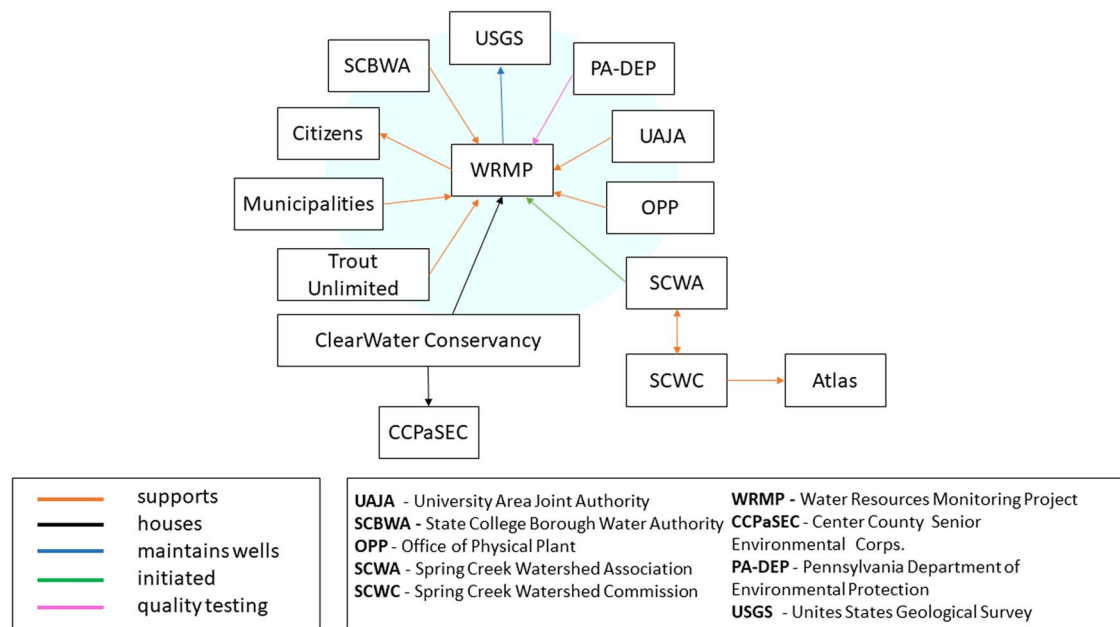


FIGURE 3. Stakeholder diagram.

a macro-invertebrate count twice a year. Their main focus is to monitor baseline water quality in the watershed. CCPaSEC has identified sites in the streams in the county's main watersheds, and teams of three to five members are responsible for monthly collections of physical site conditions, perform chemical tests and record biological data.

One overarching goal for community groups to collect and analyze water quality data is to keep a check on pollution and general health of the watershed. For instance, CCPaSEC sends their water quality data for sites that are proximal to Marcellus Shale drilling operations to a nearby university for analysis to check for problematic consequences of fracking in that region. Spring Creek Chapter of TU collects data on redd count in the Spring Creek Watershed to monitor the health of the streams and fish population since their main focus is restoration of cold-water fisheries. CCPaSEC and CWC's databases house contemporary *and* historical water quality data from the Spring Creek Watershed. By 'contemporary' we mean that these groups collect water quality data on a regular basis. However, they do not update their databases with new value measurements in real time. In some cases, it might be several days or a few weeks before newly collected data appears.

A salient point to highlight is that, in spite of the network's loosely knit structure, it is a highly effective practice. As one participant explained, *'I also work with watershed associations and conservation groups throughout the region... Basically we'll set up water quality monitoring training, help identify protocols, and train them on usage of various types of equipment, provide technical support on developing their*

study design and identifying monitoring locations (P4).'' One of the groups actively recruits senior community members in collaboration with the local Office of Aging and Retired and Senior Volunteer Program. Yet another reaches out to a younger demographic including school and college students. These members are provided with training about the use of equipment to test water quality and detailed information about the water-monitoring sites.

From our interviews, we realized that members who have been recruited continue to be involved in this practice for multiple decades. As a result, they have developed a set of expertise in this practice. They manage sample water collection, its testing, data recording, cleaning and sharing with the community. Even more so, such expertise combined with new and younger people getting involved increases the sustainability of this CoP.

4.3. Water quality data are not effectively visible to the larger community

An important detail that emerged during analysis is that existing data sets seem scattered and diverse. As a result, it is difficult to get a pulse of the whole watershed system through any single dataset. For example, both CCPaSEC and CWC maintain distinct datasets, each of which includes base cation and anion, nutrient and trace element measurements. These measures are complementary given that they are taken at different field sites in the watershed. Being cognizant of the scattered nature of the data, individual groups within the network make efforts to make this available to the public. CWC, for instance,

provides databases, educational outreach programs and other resources to support citizens and community leaders in sustainable decision-making with regard to water use and land development. CCPaSEC is concerned with raising awareness of the importance of clean water and about how the management of natural resources impacts water quality in general, both of which contribute to a broader project of promoting sustainable decision-making by citizens and community leaders. This might be one reason why they take a direct sharing approach to their data. Their data are stored in multiple Google Sheets that are visible to anyone with Internet access. However, few people seem to know that the data exist, let alone that they are publicly accessible. Other groups also maintain public databases that are more visible.

The State EPA uses water quality data to create reports and visualizations, which serve as key sources of information available to the general public about water use and water quality. The EPA provides reports for water use, but the reports are generated from various third-party sources including public water supply agencies, hydropower facilities and individuals who use $\geq 10\,000$ gallons of water per day in one month. The reports also cover various types of water use, such as drinking, industry, mining, public water supply, oil and gas, among others. The EPA utilizes GIS (Geographic Information System) for water data visualization. For example, one map depicts all the stations in the water quality network on a map of the state. They also use GIS to map impaired surface water that are not used as water supply and also the ones after implementation of a pollution control plan. EPA also makes decisions such as water body categorization and impairment identification based on their data analysis. EPA uses data to assess streams and other bodies of water to identify if they have met the standards established for water uses including aquatic life, water supply, recreation and fish consumption.

Current efforts to make information on water quality available to the public have assumed many forms. One participant mentioned, *'It's all lots of little things. For example I live near State College. My town has its own newsletter to its community. The newsletter comes out like four times a year. I write little article about like, you know, why deep rooted vegetation filters out pollution, about why you should not wash your car on the driveway, how it is not supposed to be allowed. So I'm always writing something to educate community.'* Another participant stated, *'... we have published some papers here and there. It's not really a central part of what we do. But we are involved with agencies and universities in helping them facilitate, you know, to support that sort of thing [research] (P4).'*

The hope is that these articles are consumed by community members to get a richer narrative of their local water infrastructure. However, members of the water quality community are unsure about the extent to which the broader community engages with such documents. *'We put out [a report] every year. And they are all available online too. Not sure if anybody ever reads these but the people who are actually involved in the*

watershed do, I think (P1).' So, on the one hand, community members may not be aware of these annual reports. On the other hand, they may not have the necessary literacies for collating data from multiple sources and making sense of them to understand implications of proposed developments or other changes in the community.

The WRMP has initiated a project called the Atlas, which it describes as a 'digital coffee table book' detailing the history of the watershed, summarizing pertinent issues for watershed management and collating educational materials that could be used to engage both young and adult learners on the topic of water quality. The WRMP and other stakeholders groups are interested in developing community capacity through such efforts. One participant mentioned *'We have set up a program with the local school districts so that we are part of their environmental education program. So they have classroom work they do and then they do a bus tour from here they go to the water authority (SCBWA), they go to the solid waste authority (UAJA). So they spend about a week doing classroom work and the tour of the facility (P11).'* Educational outreach in local middle and high schools, informal learning through volunteer opportunities and the watershed Atlas are just some of the ways they attempt to develop this capacity. An ideal outcome of these projects would be an increased sense of community collective self-efficacy when it comes to dealing with possible local water crises.

5. DISCUSSION

We have reported on the study of the Spring Creek Watershed involving interviews, observations and workshops with the diverse group of citizens and specialists who gather and analyze water quality data to help manage and protect this watershed. We characterized this collection of stakeholders and groups and their water quality practices as a sociotechnical system that has developed water quality monitoring practices as a form of community engagement.

The water system considered as a sociotechnical system is interesting in part because it is socially overloaded. It does not play a single role in the community but instead embodies several critical roles: (1) it is hyperlocal/extreme citizen science; monitoring and ensuring water quality involves data gathering and analysis. It documents and enriches the community's understanding of itself and provides opportunities for citizens to learn and contribute to local science. (2) It serves community welfare directly by monitoring the quality of the water residents drink every day. In this it is like volunteer fire companies, food banks and volunteers housing projects. It protects citizens from threats to local water quality that otherwise might go unnoticed. It helps citizens take more control and have more voice in managing and ensuring their own water. (3) It provides a physically active and socially rich experience to a community of (mostly) older adults. The participants in the network of community groups are hiking in woods, managing

equipment and gathering samples and observations. After a site visit, which might involve more than an hour of vigorous exercise, the groups often relax for social lunch together. The water quality monitoring activity is a vivid contribution to the community for the people who engage in it. It helps to keep them emotionally and physically healthy.

5.1. The watershed sociotechnical system is *hyperlocal* extreme citizen science

In this domain the citizens are primary stakeholders in the data: the data describes their community, they gather those data and they manage the data. This initiative is also quite consequential both in particular and as a model. Our study helps to identify a new problem area for HCI. The watershed sociotechnical system is an example of extreme citizen science (Stevens et al., 2014) in that it is much more complex and ambitious with respect to the roles of citizens. Citizens define, manage and carry out many of the core activities; their guidance and participation in water-monitoring data provides bottom-up supervision of water system management.

However, we argue that this system can also be understood as an example of *hyperlocal* (Carroll et al., 2015a) extreme citizen science that is distinguished by its focus and emphasis on local phenomena, which, in this case, pertain to water resources. It is possible, for example, for citizen scientists to contribute to broader projects, some of which may not be pertinent to local community (e.g. a study of air quality in the Northeastern United States of America, protein folding and so on), but this is not the case with the water quality practices we are studying. They are only concerned with a local watershed, and, while this watershed is part of a larger water system, and, thus, local water data has system-level implications, the motivation and intent is to understand and manage *local* water resources. These practices could also be construed as hyperlocal in the sense that each water group aligns their data practices with those of their group, which, for example, could account for different QAQC practices.

5.2. There are opportunities to strengthen coordination between groups

Interviews and workshops that were conducted have provided opportunities to discuss best practices and identify pain points with regard to different practices. For example, during an interview, we learned that two groups had developed school curricula to teach students in middle and high school about water resources. However, neither group seemed aware of the others' activities. Yet this struck us as a potential opportunity for collaboration. The groups could coordinate and support one another in developing these curricula. Sharing resources between groups is important both for public outreach initiatives and for water quality data.

For example, some groups need data from others to make decisions about how to respond to proposed regulations or land development. While data stored in a Google Doc is readily accessible by request, there are other cases where data is kept in a closed database and require some editing/cleaning before they can be released to another group. Hence, sharing data can be a time-intensive process. So far, the only solution to this problem is to make a data request in advance of any deadline for decision-making so that it arrives in time.

A significant possible community contribution of our proposed federated data platform would be a reduction in the amount of time it takes to access data that has been collected by another group. That is, data access could be made more efficient. Efficiency matters since, in many cases, water quality stakeholders work under severe time constraints, especially when it comes to local activism. Rather than make a request, for instance, a stakeholder could visit the data platform to curate and download the data they need ad hoc, which is typical of open data platforms. However, this efficiency access comes at a cost. Currently, data sharing within the local community is a mixture of independent and collaborative processes. There are some open repositories accessible without request while others, that, based on our stakeholder interviews, are more comprehensive, are accessible only through a 'gatekeeper', who is also a member of the water quality community. These closed repositories are accessible by request. One person makes a request to another person for data, and that second person fulfills the request.

An open, federated data platform reduces the need for stakeholders to interact with one another when it comes to data sharing, and this could have an adverse effect on the water quality community. For example, requesting data from another stakeholder can make activities within the water quality community more visible; one person can get a sense for what another is working on based on the data they are requesting. At the very least, a data request creates the potential for conversations about current water quality projects. The requester may include a brief description of their project as part of their request. Alternatively, the person fulfilling the request may ask about the project. Thus, we need to think about what it would mean to reduce the potential for conversations and, potentially, to do away with certain roles in the community. It could be the case that efficiency and access are not worth these costs.

On the other hand, while the potential for these interactions exist, there is little evidence that they are already taking place. For example, data requests to WRMP can be made through a web form (link removed for blind review). The form does not require a rationale or project description, and it states that the entire WRMP database is 'free and available upon request'. It neither encourages nor discourages richer conversations between stakeholders. Data requests to WRMP constitute a formal collaboration between groups—it is an established and accepted aspect of data sharing. Richer conversations between

the requester and the requestee could be construed as *potential* informal collaborations, since we do not yet know whether such conversations take place. How might such informal collaborations motivate (or not) stakeholders to participate? If data requests are fulfilled without any additional interaction between stakeholders, then they may not be a source of additional motivation to engage in watershed stewardship. This, in turn, could mean that an open, federated data platform—one that circumvents the need to make data requests to a human stakeholder—might not have significant adverse effects on engagement.

5.3. Water quality practices can be made more visible

Rather than inventing or trying to get people to adopt a critical collective practice, we discovered that the practice already exists. The rich foundation of water quality practices and expertise has helped the local community to take more control of watershed management and planning. The network of community groups that were identified now work with involves dozens of active participants. The majority are volunteers, but, distinct from other volunteer initiatives, this social system appears to be stable and productive. The water quality volunteers are active and engaged with the larger community. Older adults, for example, enjoy a sense of community and purpose in their participation, and they engage in health co-production by traveling out to field sites to collect water quality samples, socializing with other water quality volunteers and participating in local governance. However, water quality monitoring and water data have not been recognized by most of the community as a typical way to engage in community life. The potential to enrich and to help the community remains under leveraged.

Collaborative community water data can strengthen communities in several key respects beyond safeguarding water supplies. First, water data are consequential to community members, and thus provide motivation for informal learning and development of data analytic concepts and skills. Data literacy is increasingly important for participation in citizenship and everyday life. Second, water data can provide factual grounding for creative community discussions and deliberation in civic contexts, ranging from public hearings about re-zoning and land development to casual conversations with colleagues and neighbors. The use of data and data-driven argumentation to shape decisions and policies at the local level provides opportunity to strengthen local democracy and democratic practices in general. Finally, participation in water monitoring and data analysis, and in public discussion and interpretation of water data, evolves and opens a key topic of contemporary municipal governance to those with commitment and expertise. Data analytics and visualization can provide foundation and focus for community innovation; enabling concrete possibilities for enhancing community learning, civic participation and municipal governance.

Water quality boundary objects are emerging that can integrate and ground discussions of water quality throughout the community in the form of the Atlas and our proposed federated water data platform. Along with our community partners, we envision the data platform as a public system that could become a community resource for informal learning about water data and natural resource management and a starting point for broader citizen engagement and participation (Bonney *et al.*, 2014). Together with members of the water quality community, we are investigating the role the data platform plays in making local water quality data more visible to citizens, in evoking broader engagement in environmental stewardship and in enabling citizens and municipal leaders to undertake more informed and effective policy and management regarding water resources.

6. FUTURE WORK

We believe that understanding how the sociotechnical water system of Spring Creek produces high quality water can serve as a paradigm case to, for example, (1) guide broader research into awareness and engagement of citizens with local water systems in the United States, (2) develop a dissemination intervention to evoke more effective citizens engagement with water systems, and (3) provide specific citizen-oriented requirements for emerging water data products and platforms.

6.1. Surveying citizen awareness and engagement

Based on our initial description of the Spring Creek Watershed sociotechnical system, we see value in carrying out a large-scale survey study to investigate what Americans know about their local water supplies, including concerns and desires to know (and engage) more. The NWQMC (2018) open data repository provides the most comprehensive resource for water quality and demographics of local populations, but it includes no data regarding engagement, knowledge or attitudes of local populations with respect to water quality. The survey could focus on three items: (1) local water quality awareness, (2) interest in local water monitoring and water data and (3) general community constructs such as attachment, engagement, social support networks, sense of community and community collective efficacy.

6.2. Translational hyperlocal data manual

We will design and develop a translational hyperlocal data manual, which will serve as a resource for citizens and local governments. Such a manual could serve as a framework for creating and evaluating water quality data implementations in local communities using success benchmarks that we identify, much like the STAR Communities (<http://www.starcommunities.org>) sustainability planning framework. We propose to demonstrate the value and utility of developing water quality data literacy

and provide illustrative, interactive case studies that people can engage with in order to learn more about data, data analysis and, crucially, translate the outcomes of analysis into civic participation. Although there are many general resources people can use to learn about data (e.g. existing open data platforms) and data analysis (e.g. Khan Academy) there are fewer resources for citizens who are interested in translating what they learn about data to local civic participation. There is a real need for translational resources. The creation of a manual could also serve as a model for other communities to develop their own translational hyperlocal data resources to promote data-driven, civic participation.

6.3. Design requirements for water data platforms

Open data platforms exist to provide access and instructional support for working with different kinds of data. These platforms are shared resources, and, over the years, various sets of design principles have been proposed to help guide designers in the creation of new shared resource platforms. The most famous set is Elinor Ostrom's eight design principles for long-enduring common property resource institutions (Ostrom, 1990), which can be applied to open access resources. Ostrom's eight principles generalize some of the characteristics that shared resources need to possess if they are to be sustainable. However, the principles are limited in what they can tell us about unique local needs. In other words, while many long-enduring shared resources embody all eight principles, they also possess unique characteristics based on local needs. This has implications for how designers go about creating new shared resource artifacts and systems. As part of this project, we see opportunities to develop sets of methods to support designers in identifying unique local needs, so that novel systems and artifacts can be designed with both generic principles for sustainability and unique local ones in mind.

7. CONCLUSIONS

The recent disasters in Flint and Cape Town are both failures at various levels of government planning and management. In both cases, community engagement started when the disasters had already occurred. The human race needs to do better, and our work has identified a working example as a starting point.

In this paper, we showed how local water quality groups can organize to gather and analyze water data on behalf of their fellow citizens. The groups employ scientific methods and instruments; they learn about and carry out hyperlocal citizen science, that is to say, water quality science that they plan and manage, and that is focused on their own local public water supply to shape local public policy and ensure the future of local water. This practice is a kind of community engagement analogous to food banks and emergency services, neighbors take care of neighbors by protecting the water supply they all depend upon.

Our analysis also emphasized challenges and opportunities for citizen water quality monitoring, especially digital infrastructure and tools to enhance coordination and visibility within and among the groups, and with regard to the larger community. The citizens we worked with clearly cherish the autonomy of their individual groups and find meaning in the data they gather, but they also clearly wish to share their data with others, and to see and use a wider range of the community's data. Data management strategies and digital tools that facilitate both requirements could strengthen the effectiveness of the individual groups and the overall watershed community while at the same time supporting their core values.

The water quality practice is often invisible community work: much of it is carried out at a stream in the deep woods, or in a laboratory. Even the other water groups are only vaguely aware of specific contributions, members of the larger community have little awareness of this work at all. In part, this is paradigmatic of invisible work; the people doing the work are not powerful, and their activities are not headlines in the community. If everything is fine with local public water, then there is no news to be aware of. Nonetheless, it might be better for the larger community and for the water groups for everyone to be a little more aware of water quality monitoring, both because awareness of water quality is part of the collective capacity to ensure water quality, and because public awareness of community service helps encourage more engagement by more people throughout the community.

Water quality monitoring and data analysis is community work. Such work protects the future of local water, a key resource for a viable community and a resource no one can take for granted. A resilient and scalable approach to such work is to engage community members themselves in organizing and carrying this work out. Indeed, we found that this approach is already highly developed and stabilized in our community. It involves many groups and individuals, and an extensive network of organizational dependencies and working relationships, and member bridging.

These groups, including their expertise, equipment, practices and data, are a huge resource to the community that can be better understood and supported. The groups and individuals we worked with are very interested in how collaborative systems and services could support them and make them more effective to the larger community. Re-framing water quality as collaborative community work is a transformational idea that is obvious but neither well understood nor fully developed and exploited. We suggest that this is a way forward not only for water quality but also for HCI. Water quality is a global crisis, and humans are the key.

Our study showed how HCI concepts and techniques can help us identify and understand community practices and possibilities in a complex and highly consequential activity context. It suggests directions for broadening HCI's foundational concept of 'context of use' toward sustainable, scalable and effective sociotechnical collaboration.

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A. Appendix

Interviews	
Participants	Water group
P1	CWC
P2	TU
P3	UAJA
P4	TU
P5	PA-FBA
P6	TU
P7	NVEC
P8	SCWC
P9	SCWC
P10	OPP
P11	UAJA
P12	NVEC
P13	CCPaSEC
P14	CCPaSEC
P15	SCBWA
Hackathons	
Hackathons	Water groups
H1	Water stakeholders (WRMP, CWC, TW, PaSEC, municipality)
H2	Water stakeholders (CWC, CCPaSEC, SCWA, Atlas, UAJA)
H3	Water stakeholders (UAJA, SPWC, municipality, public library)